

Partha Pratim Deka

Joint Postdoctoral Researcher · UCT & SARAO

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Professional Appointments

Joint Postdoctoral Researcher

03/2025 – present

University of Cape Town (UCT) & South African Radio Astronomy Observatory (SARAO)

Role: Astronomer on Duty

Education

Doctor of Philosophy (Ph.D.)

08/2019 – 02/2025

The Inter-University Centre for Astronomy and Astrophysics (IUCAA), Pune

Thesis: An unbiased view of cold atomic gas associated with radio loud AGNs

Thesis Advisor: Professor Neeraj Gupta

Master of Science (M.Sc.), Physics & Astrophysics

2016 – 2018

Hansraj College, University of Delhi, Delhi, India

Bachelor of Science (B.Sc.), Physics (Honors)

2013 – 2016

Ramjas College, University of Delhi, Delhi, India

Research Interests

› Data intensive astronomy

- › Radio survey planning, catalog generation and characterization
- › Multi-wavelength characterization of radio sources
- › Applications of machine learning in astronomy

› Galaxy evolution

- › Kinematics and distribution of cold gas in galaxies and its redshift evolution
- › Active Galactic Nuclei (AGN) feeding and feedback
- › High redshift radio galaxies

Publications

Lead Author Publications

- [1] P. P. Deka, N. Gupta, and MALS collaboration. “Search for neutral *HI* 21-cm absorption associated with low- and high-excitation radio-loud AGNs from MALS”. Under review.

- [2] P. P. Deka, N. Gupta, H. W. Chen, S. D. Johnson, P. Noterdaeme, F. Combes, E. Boettcher, S. A. Balashev, K. L. Emig, G. I. G. Józsa, H. -R. Klöckner, J. -K. Krogager, E. Momjian, P. Petitjean, G. C. Rudie, J. Wagnveld, and F. S. Zahedy. "MALS discovery of a rare H I 21 cm absorber at $z \sim 1.35$: Origin of the absorbing gas in powerful active galactic nuclei". In: *A&A* 687, A50 (July 2024), A50. DOI: [10.1051/0004-6361/202348464](https://doi.org/10.1051/0004-6361/202348464). arXiv: 2311.00336 [astro-ph.GA].
- [3] P. P. Deka, N. Gupta, P. Jagannathan, S. Sekhar, E. Momjian, S. Bhatnagar, J. Wagnveld, H. -R. Klöckner, J. Jose, S. A. Balashev, F. Combes, M. Hilton, D. Bargaonkar, A. Chatterjee, K. L. Emig, A. N. Gaunekar, G. I. G. Józsa, D. Y. Klutse, K. Knowles, J. -K. Krogager, A. Mohapatra, K. Moodley, Sébastien Muller, P. Noterdaeme, P. Petitjean, P. Salas, and S. Sikhosana. "The MeerKAT Absorption Line Survey (MALS) Data Release. I. Stokes I Image Catalogs at 1–1.4 GHz". In: *ApJS* 270.2, 33 (Feb. 2024), p. 33. DOI: [10.3847/1538-4365/acf7b9](https://doi.org/10.3847/1538-4365/acf7b9).
- [4] P. P. Deka, G. C. Dewangan, K. P. Singh, and J. Postma. "A pair of UV nuclei or a compact star-forming region near the active nucleus in Mrk 766?" In: *Journal of Astrophysics and Astronomy* 42.2, 46 (Oct. 2021), p. 46. DOI: [10.1007/s12036-021-09695-6](https://doi.org/10.1007/s12036-021-09695-6). arXiv: 2101.07537 [astro-ph.GA].

Co-authored Publications

- [1] N. Gupta, J. Kerp, S. A. Balashev, A. P. M. Morelli, F. Combes, J.-K. Krogager, E. Momjian, D. Bargaonkar, P. P. Deka, K. L. Emig, J. Jose, G. I. G. Józsa, H.-R. Klöckner, K. Moodley, S. Muller, P. Noterdaeme, P. Petitjean, and J. D. Wagnveld. "The MeerKAT Absorption Line Survey (MALS) data release 3: Cold atomic gas associated with the Milky Way". In: *A&A* 698, A120 (June 2025), A120. DOI: [10.1051/0004-6361/202452407](https://doi.org/10.1051/0004-6361/202452407). arXiv: 2504.00097 [astro-ph.GA].
- [2] Morgan Himes, Preshanth Jagannathan, Dale A. Frail, Frank Schinzel, Neeraj Gupta, S. A. Balashev, F. Combes, P. P. Deka, H. -R. Klöckner, E. Momjian, and J. D. Wagnveld. "Fermi Unassociated Sources in the MeerKAT Absorption Line Survey". In: *ApJ* 980.1, 81 (Feb. 2025), p. 81. DOI: [10.3847/1538-4357/ada1cf](https://doi.org/10.3847/1538-4357/ada1cf). arXiv: 2412.12470 [astro-ph.HE].
- [3] F. Combes, N. Gupta, S. Muller, S. Balashev, P. P. Deka, K. L. Emig, H. -R. Klöckner, D. Klutse, K. Knowles, A. Mohapatra, E. Momjian, P. Noterdaeme, P. Petitjean, P. Salas, R. Srianand, and J. D. Wagnveld. "PKS 1413+135: OH and H I at $z = 0.247$ with MeerKAT". In: *A&A* 671, A43 (Mar. 2023), A43. DOI: [10.1051/0004-6361/202245482](https://doi.org/10.1051/0004-6361/202245482). arXiv: 2211.09355 [astro-ph.GA].
- [4] Kimberly L. Emig, Neeraj Gupta, Pedro Salas, Sébastien Muller, Sergei A. Balashev, Françoise Combes, Emmanuel Momjian, Yiqing Song, Preshanth Jagannathan, Partha P. Deka, Gyula I. G. Józsa, Hans-Rainer Klöckner, Abhisek Mohapatra, Pasquier Noterdaeme, Patrick Petitjean, Raghunathan Srianand, and Jonah D. Wagnveld. "Discovery of Hydrogen Radio Recombination Lines at $z = 0.89$ toward PKS 1830-211". In: *ApJ* 944.1, 93 (Feb. 2023), p. 93. DOI: [10.3847/1538-4357/acb49d](https://doi.org/10.3847/1538-4357/acb49d). arXiv: 2301.03115 [astro-ph.GA].
- [5] J. D. Wagnveld, H. -R. Klöckner, N. Gupta, P. P. Deka, P. Jagannathan, S. Sekhar, S. A. Balashev, E. Boettcher, F. Combes, K. L. Emig, M. Hilton, G. I. G. Józsa, P. Kamphuis, D. Y. Klutse, K. Knowles, J. -K. Krogager, A. Mohapatra, E. Momjian, K. Moodley, S. Muller, P. Petitjean, P. Salas, S. Sikhosana, and R. Srianand. "The MeerKAT Absorption Line Survey: Homogeneous continuum catalogues towards a measurement of the cosmic radio dipole". In: *A&A* 673, A113 (May 2023), A113. DOI: [10.1051/0004-6361/202245477](https://doi.org/10.1051/0004-6361/202245477). arXiv: 2302.10696 [astro-ph.CO].

- [6] N. Gupta, G. Shukla, R. Srianand, J. -K. Krogager, P. Noterdaeme, A. J. Baker, F. Combes, J. P. U. Fynbo, E. Momjian, M. Hilton, T. Hussain, K. Moodley, P. Petitjean, H. -W. Chen, P. Deka, R. Dutta, J. Jose, G. I. G. Józsa, C. Kaski, H. -R. Klöckner, K. Knowles, S. Sikhosana, and J. Wagnveld. "MALS SALT-NOT Survey of MIR-selected Powerful Radio-bright AGN at $0 < z < 3.5$ ". In: *ApJ* 929.1, 108 (Apr. 2022), p. 108. doi: 10.3847/1538-4357/ac4220. arXiv: 2107.09705 [astro-ph.GA].
- [7] E. K. Maina, Abhisek Mohapatra, G. I. G. Józsa, N. Gupta, F. Combes, P. Deka, J. D. Wagnveld, R. Srianand, S. A. Balashev, Hsiao-Wen Chen, J. -K. Krogager, E. Momjian, P. Noterdaeme, and P. Petitjean. "Mapping H I 21-cm in the Klemola 31 group at $z = 0.029$: emission and absorption towards PKS 2020-370". In: *MNRAS* 516.2 (Oct. 2022), pp. 2050–2061. doi: 10.1093/mnras/stac1752. arXiv: 2206.09940 [astro-ph.GA].

Mentorship & Student Supervision

Co-Supervisor, Master's Student

01/2026 – present

University of Cape Town

Project: Detection and characterization of radio sources from MeerLIRGs and finding anomalous radio sources using Machine Learning algorithms.

Skills

Communication	English (fluent), Hindi (fluent), Assamese (native).
Language	PYTHON, C, C++, SQL
Software	CASA, Cube Analysis and Rendering Tool for Astronomy (CARTA), Python Blob Detector and Source Finder (PyBDSF), Aladin, TOPCAT, DS9, Photoionization simulation code CLOUDY, Astronomical SURVival Statistics (ASURV), Galaxy/AGN Emission-Line Analysis Tool (GELATO), Galaxy SED model codes (BAGPIPES, CIGALE)
Data Handling	Skilled in the analysis of Astronomical data, with a primary focus on radio data.
Misc.	Experienced in working with operating systems including macOS, Linux and Windows as well as text editing with TEX.

Awards and Achievements

-  **2021–2025:** Senior Research Fellow, IUCAA, Pune.
-  **2019–2021:** Junior Research Fellow, IUCAA, Pune.
-  **2018–2019:** Junior Research Fellow, Physical Research Laboratory (PRL), Ahmedabad.
-  **2018:** Joint CSIR-UGC Junior Research Fellowship & Eligibility for Lectureship (NET-JRF), June 2018, All India Rank (AIR)-89.
-  **2018:** Graduate Aptitude Test in Engineering (GATE), AIR-227.
-  **2017–2018:** Meritorious Award in Masters for ranking among Top 10.
-  **2013–2018:** INSPIRE scholarship for scoring top 1% in Senior secondary examination.

Certification

-  **2023:** Supervised Machine Learning: Regression and Classification. Authorized by DeepLearning.AI and Stanford University and offered through Coursera.

Invited Talks & Colloquia

-  **October 2025:** *Toward an Unbiased View of Cold Gas in Radio-Loud AGNs*, Colloquium delivered at the South African Astronomical Observatory (SAAO), Cape Town, South Africa.

Schools, Workshops and Conferences Attended

-  **May 2026:** *An unbiased view of cold-atomic gas associated with radio-loud AGNs*, poster presented at the 44th annual meeting of Astronomical Society of India (ASI).
-  **June 2025:** *An unbiased view of cold-atomic gas associated with radio-loud AGNs*, poster presented at the conference 'A new era in astrophysics: Preparing for early science with the SKAO' held in Görlitz, Germany.
-  **December 2024:** *An unbiased view of cold atomic gas associated with radio-loud AGNs in MALS*, contributed talk (in-person) at the conference 'Baryons Beyond Galactic Boundaries – 2024', held at IUCAA, Pune.
-  **May 2024:** *MALS Data Release I: Probing Evolution of Cold gas in AGNs*, contributed talk (in-person) at the conference 'SPARCS XII : Pushing Towards the Final Frontier', held in Bologna, Italy.
-  **September 2023:** *MALS sample of radio-loud AGNs: Signatures of jet-cloud interaction*, contributed talk (in-person) at the conference 'AGN on the beach' held in Tropea, Italy.
-  **August 2023:** *MALS: results from continuum and spectral line data products*, contributed talk (online) at the Asia-Pacific Regional IAU Meeting (APRIM), Koriyama, Japan.
-  **March 2023:** *MALS: Radio continuum catalog at 1.0 and 1.4 GHz*, contributed talk (in-person) at the 41st annual meeting of Astronomical Society of India (ASI).
-  **March 2022:** *Preparing for an unbiased view of cold atomic gas associated with radio loud AGNs up-to $z \sim 2$ using MALS*, poster (in-person) at the 40th annual meeting of Astronomical Society of India (ASI).

Teaching Assistant & Science Popularization

-  **2024:** Member of Scientific Organizing Committee (SOC) for the Young Astronomers' Meet (YAM), CHRIST University, Bangalore, India.
-  **2020–2023:** Volunteer, National Science Day at IUCAA, Pune, India.
-  **2023:** Tutor for the 14 lecture course on Interstellar medium by Prof. N. Gupta as part of the IUCAA-NCRA Graduate school.
-  **2023:** Organized a hands-on session on radio astronomy data handling during the Introductory Summer School in Astronomy & Astrophysics (ISSAA-2023).
-  **2022:** MALS Unveiled: Revealing Half a Million Cosmic Radio Sources, contributed article to CosmicVarta.
-  **2022–2024:** Co-chaired MALS Working Group-1 (WG1): Cold atomic and molecular gas absorption Galactic/Extragalactic - along with Prof. Françoise Combes.